

Scenario: Smart Utility Billing System

A city council has introduced a **Smart Utility Billing System** to calculate monthly electricity bills for households. The system must automatically compute charges based on electricity usage, apply discounts, and include penalties where necessary.

Each household provides the following details:

- Customer Name
- Units of electricity consumed (in kWh)

The billing rules are as follows:

1. The cost per unit is:
 - First 100 units → P1.50 per unit
 - Next 200 units → P2.00 per unit
 - Above 300 units → P3.00 per unit
2. Additional rules:
 - If total units exceed 350, apply a **5% surcharge** on the total bill.
 - If total units are less than 100, apply a **10% discount**.
 - A fixed service charge of P50 is added to every bill.

Tasks

Task 1: Input (10 marks)

Write a Python program that prompts the user to input:

- Customer name
- Number of units consumed

Task 2: Bill Calculation (25 marks)

Using appropriate control structures:

- Calculate the total cost based on the tiered pricing system
- Apply surcharge or discount where applicable

- Add the fixed service charge

Task 3: Output (10 marks)

Display the following neatly:

- Customer name
- Units consumed
- Total bill before adjustments
- Discount or surcharge applied
- Final amount to be paid

Task 4: Mathematical Analysis (15 marks)

Enhance the program to:

- Calculate and display the **average cost per unit**
- Round all monetary values to **2 decimal places**

Task 5: Extension (10 marks)

Modify the program to process **multiple customers (at least 3)** and:

- Store their final bills in a list
- Display the **highest and lowest bill amounts**